Ricardo Castro

rickycastro2023@gmail.com | sites.google.com/view/ricardocastro

Education

Carnegie Mellon University, Pittsburgh, PA — GPA: 3.90
BS in Mechanical Engineering — Expected May 2027
MS in Mechanical Engineering — Expected May 2027
Coursework: Optimal Control & RL (graduate-level), AI/ML for Manufacturing (graduate-level), Dynamic Systems and Controls

Experience

Mechanical Design Intern, Origami Robotics – Pittsburgh, PA — August 2025 – Present

- Designed a glove controller in Onshape that maps potentiometer rotation to servo-driven joints
- Customized 3D-printed finger joints to individual users, reducing complexity with print-in-place linkages
- Optimized linkage geometry via motion analysis for prototype showcased in succesful seed funding demos

Robotics Research Assistant, Robomechanics Lab – Pittsburgh, PA — Apr 2025 – Present

- Scaled a miniaturized bipedal robot to human size by redesigning geometry and mass properties in Onshape
- Simulated full 3D locomotion in MuJoCo with state estimation and tuned PD control, adjusting mechanical design via reward-based evolutionary strategies for gait optimization
- Assembled carbon fiber and 3D-printed limb prototypes for hardware validation
- Built a Raspberry Pi motor interface over CAN to gather motion data, refining control system to achieve walking

Manufacturing Intern, Manufacturing Futures Institute – Pittsburgh, PA — May 2024 – Aug 2024

- Programmed MIG and laser welding workflows on the Lincoln Electric Classmate Laser, creating a repeatable demonstration through FANUC Controller
- Created a modular fixturing table with a tiltable frame up to 30 degrees in SolidWorks and machined components via HAAS CNC and water jet to balance structural stability with accessibility

Projects

Greek Sing Sets Chair — August 2024 – Present

- Led the design and construction of a 20’ × 12’ modular stage set with platforms, doors, and moving elements
- Modeled structures in SolidWorks and led 200+ fabrication hours with wood cutting tools to deliver builds under time and safety constraints

Motor Controlled Scissors — sites.google.com/andrew.cmu.edu/marketproject/

- Created a motorized scissor system to reduce joint strain for users with tendinitis, combining electromechanical control and ergonomic design for accessible one-handed cutting
- Modeled an electromechanical cutting system in MATLAB and Simulink to tune PID gains under ramp inputs
- Implemented and validated a PID-controlled motor system on Arduino to ensure consistent blade closure across varying loads and user-triggered interruptions

Rethink the Rink — sites.google.com/view/ricardocastro

- Led material ideation, CAD modeling, and prototyping across two hockey safety challenges, developing protective glove and neck guard systems in collaboration with Covestro and the Pittsburgh Penguins
- Recognized with Most Innovative Design (2024) and Team MVP (2025)

Skills

Languages: English (Native), Spanish (Fluent)
Programming & Simulation: Python, MATLAB, Simulink, C++, MuJoCo, Gazebo, ANSYS (FEA)
Design & CAD: SolidWorks, Onshape, Fusion 360, AutoCAD, DFM/DFAM, Technical Drawings, GD&T
Fabrication & Prototyping: CNC Machining, 3D Printing, Laser Cutting, MIG/Laser Welding, Manual Milling